

# Generation Interconnection System Impact Study Report

## For Queue Project Q15-0470

Hard Rock 138 kV

March 22, 2017

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0470. The Interconnection Customer proposes a 1056.2 MW natural gas combined cycle facility interconnecting at the Hard Rock 138 kV substation in the Virginia area. The request was received on May 01, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 203 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$187,833,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                         | Contingency                                  | Loading (%) | Rating (MVA) | DFAX (%) |
|--------------------------------------------|----------------------------------------------|-------------|--------------|----------|
| Canon City-Hard Rock 138 kV                | Hard Rock-East Longmeadow 138 kV             | 473.0       | 223          | 100.0    |
| E Longmeadow-Hard Rock 138 kV L            | Canon City-Hard Rock 138 kV                  | 468.6       | 225          | 100.0    |
| Villa Del Rey 3010-Canon City 138 kV       | Hard Rock-E Longmeadow 138 kV                | 314.9       | 209          | 51.5     |
| Deen Point-Gldfield Substation 138 kV      | North Tower Tap-Mccartney 345 kV             | 186.3       | 214          | 16.8     |
| Sanger-Cochise 138 kV line                 | Canon City-Hard Rock 138 kV                  | 186.0       | 255          | 32.6     |
| Tyler-Kewaunee 138 kV                      | Mccartney-N Tower Tap 345 kV                 | 165.0       | 227          | 16.8     |
| Canon City-Seen Mile 138 kV                | Hard Rock-E Longmeadow 138 kV line           | 160.8       | 242          | 48.5     |
| Moonstone-Deltaport 138 kV                 | Haggin-Tyler 138 kV Line                     | 157.8       | 222          | 9.6      |
| Seven Mile-Rosemary Power Sta 138 kV line  | E Longmeadow-Hard Rock 138 kV                | 154.3       | 245          | 48.5     |
| Wagley Robertson-Disputanta 138 kV         | Floraday-Sonoma 3 138 kV                     | 151.7       | 170          | 13.5     |
| Kanaka-Youngcane 138 kV                    | Mccartney-North Tower Tap 345 kV             | 150.5       | 156          | 8.3      |
| Bobville-Cochise 138 kV                    | Canon City Substation-Hard Rock 138 kV line  | 149.5       | 317          | 32.6     |
| Cedar Bayou Plant-Jim Hill 138 kV          | Skaggs Island Junction 2-Mitchel Dam 345 kV  | 146.3       | 236          | 6.9      |
| Cedar Bayou Plant-Centralia 138 kV         | Mitchell Dam-Skaggs Island Jct 2 345 kV Line | 146.0       | 208          | 6.9      |
| Floraday-Disputanta Substation 138 kV line | Floraday-Sonoma 3 138 kV                     | 137.3       | 202          | 13.5     |
| N Randolph-Yellow Creek 138 kV             | Hull St Rd-Haggin 345 kV                     | 136.9       | 241          | 5.3      |

| Monitored Facility                               | Contingency                              | Loading (%) | Rating (MVA) | DFAX (%) |
|--------------------------------------------------|------------------------------------------|-------------|--------------|----------|
| North Randolph-Bairoa 3008 138 kV Line           | Hull St Road-Haggin 345 kV               | 134.1       | 237          | 5.3      |
| Parsons Tap-E Longmeadow 138 kV                  | Hard Rock-Canon City 138 kV              | 133.8       | 324          | 51.8     |
| Rosemary Power Sta-Roanoke Rapids 138 kV         | E Longmeadow-Hard Rock Substation 138 kV | 132.7       | 186          | 31.2     |
| East Longmeadow-Sanger 138 kV L                  | Canon City-Hard Rock 138 kV              | 132.5       | 470          | 48.2     |
| Yarnen 345/138kV Transformer                     | Mccartney-Yarnen 345 kV Line             | 129.0       | 321          | 7.3      |
| Pacifica Substation-Westrock Souteast Llc 138 kV | Mitchel Dam-Skaggs Island Jct 2 345 kV   | 129.0       | 166          | 6.0      |
| Oyster Riv-Harvell Substation 138 kV             | N Tower Tap-Haggin 345 kV                | 128.5       | 191          | 5.2      |
| Potosi-Deltaport 138 kV                          | Tyler-Haggin Substation 138 kV           | 127.3       | 301          | 9.6      |
| Roanoke Rapids-Mccartney 345 kV                  | Roanoke Rapids-Gaston 345 kV Line        | 126.7       | 1,398        | 14.7     |

### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

## **4. Required Network Upgrades and Cost Estimates**

| Facility                                   | Required Upgrade                            | Cost Estimate | Allocation to Project |
|--------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Canon City-Hard Rock 138 kV                | Rebuild line with higher-capacity conductor | \$5,633,000   | 100% (\$5,633,000)    |
| E Longmeadow-Hard Rock 138 kV L            | Rebuild line with higher-capacity conductor | \$5,649,000   | 100% (\$5,649,000)    |
| Villa Del Rey 3010-Canon City 138 kV       | Rebuild line with higher-capacity conductor | \$28,235,000  | 100% (\$28,235,000)   |
| Deen Point-Goldfield 138 kV                | Rebuild line with higher-capacity conductor | \$2,962,000   | 39% (\$1,149,000)     |
| Sanger-Cochise 138 kV line                 | Rebuild line with higher-capacity conductor | \$2,077,000   | 94% (\$1,947,000)     |
| Tyler-Kewaunee 138 kV                      | Rebuild line with higher-capacity conductor | \$6,106,000   | 39% (\$2,369,000)     |
| Canon City-Seen Mile 138 kV                | Rebuild line with higher-capacity conductor | \$4,986,000   | 100% (\$4,986,000)    |
| Deltaport Substation-Moonstone 138 kV line | Rebuild line with higher-capacity conductor | \$1,916,000   | 48% (\$916,000)       |
| Seven Mile-Rosemary Power Sta 138 kV line  | Rebuild line with higher-capacity conductor | \$50,825,000  | 100% (\$50,825,000)   |
| Wagley Robertson-Disputanta 138 kV         | Rebuild line with higher-capacity conductor | \$11,537,000  | 93% (\$10,764,000)    |
| Kanaka-Youngcane 138 kV L                  | Rebuild line with higher-capacity conductor | \$6,298,000   | 50% (\$3,154,000)     |
| Bobville-Cochise 138 kV                    | Rebuild line with higher-capacity conductor | \$9,273,000   | 94% (\$8,696,000)     |

| Facility                                         | Required Upgrade                            | Cost Estimate | Allocation to Project |
|--------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Cedar Bayou Plant-Jim Hill 138 kV                | Rebuild line with higher-capacity conductor | \$6,738,000   | 22% (\$1,515,000)     |
| Cedar Bayou Plant-Centralia 138 kV               | Rebuild line with higher-capacity conductor | \$961,000     | 22% (\$216,000)       |
| Floraday-Disputanta Substation 138 kV line       | Rebuild line with higher-capacity conductor | \$9,959,000   | 93% (\$9,292,000)     |
| N Randolph-Yelow Creek 138 kV                    | Rebuild line with higher-capacity conductor | \$7,177,000   | 69% (\$4,947,000)     |
| North Randolph-Bairoa 3008 138 kV                | Rebuild line with higher-capacity conductor | \$5,981,000   | 69% (\$4,122,000)     |
| Parsons Tap-E Longmeadow 138 kV                  | Rebuild line with higher-capacity conductor | \$20,833,000  | 93% (\$19,304,000)    |
| Rosemary Power Sta-Roanoke Rapids 138 kV         | Rebuild line with higher-capacity conductor | \$4,131,000   | 100% (\$4,131,000)    |
| East Longmeadow-Sanger 138 kV L                  | Rebuild line with higher-capacity conductor | \$7,305,000   | 92% (\$6,732,000)     |
| Yarnen 345/138 kV XFMR                           | Replace transformer with higher-rated unit  | \$14,350,000  | 20% (\$2,933,000)     |
| Pacifica Substation-Westrock Souteast Llc 138 kV | Rebuild line with higher-capacity conductor | \$3,645,000   | 60% (\$2,200,000)     |
| Oyster Riv-Harvell Substation 138 kV             | Rebuild line with higher-capacity conductor | \$1,871,000   | 22% (\$415,000)       |
| Potosi-Deltaport 138 kV                          | Rebuild line with higher-capacity conductor | \$6,974,000   | 48% (\$3,332,000)     |
| Roanoke Rapids-Mccartney 345 kV                  | Rebuild line with higher-capacity conductor | \$47,950,000  | 9% (\$4,371,000)      |
|                                                  | Total allocated to project                  |               | \$187,833,000         |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.